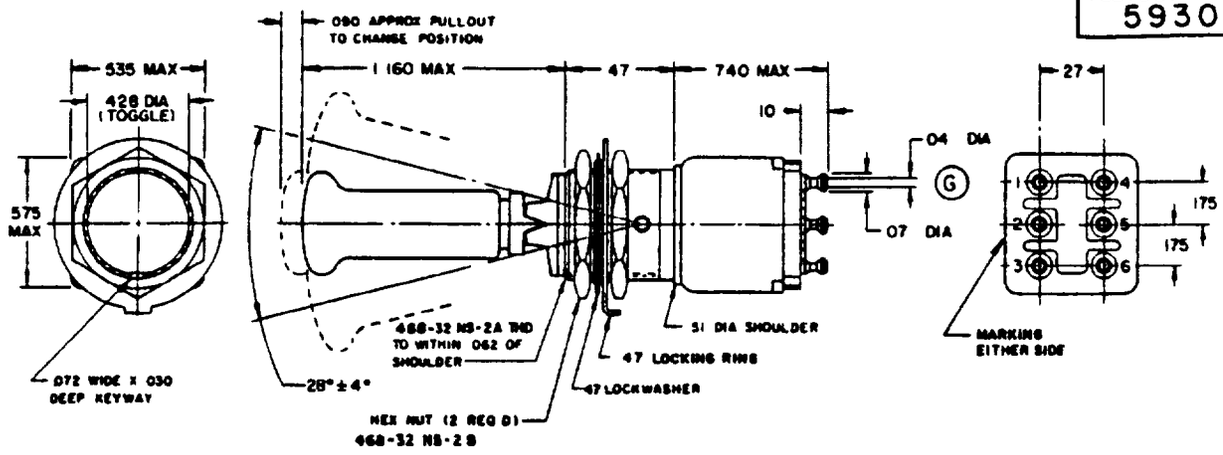


Inches	mm	Inches	mm	Inches	mm
.005	0.13	.072	1.83	.47	11.94
.020	0.51	.090	2.29	.51	12.95
.030	0.76	.10	2.54	.535	13.59
.04	1.02	.175	4.45	.575	14.61
.062	1.57	.27	6.68	.740	18.80
.07	1.78	.428	10.87	1.160	29.46

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.020 on two place decimals and ±.005 on three place decimals.
3. For hardware detail specifications, see supplement of MIL-S-83731.
4. Configuration of switch case is optional provided maximum dimensions specified are not exceeded.
5. Metric equivalents are given for general information only.

DETAIL REQUIREMENTS:

LOCKING COMBINATIONS (KEYING SIDE)

A	B	D	E	F	G	H
LOCKED IN THREE POSITIONS	LOCKED IN CENTER AND DOWN POSITION (KEYING SIDE)	LOCKED OUT OF CENTER POSITION	LOCKED IN CENTER POSITION	LOCKED IN UP POSITION (OPPOSITE KEYING)	LOCKED IN DOWN POSITION (KEYING SIDE)	LOCKED OUT OF CENTER AND DOWN POSITION (KEYING SIDE)
J	K	L	M	N	P	
LOCKED OUT OF CENTER AND UP POSITION (OPPOSITE KEYING)	LOCKED IN CENTER AND UP POSITION (OPPOSITE KEYING)	LOCKED OUT OF DOWN POSITION (KEYING SIDE)	LOCKED OUT OF AND INTO UP POSITION (OPPOSITE KEYING)	LOCKED OUT OF UP POSITION (OPPOSITE KEYING)	LOCKED OUT OF AND INTO DOWN POSITION (KEYING SIDE)	FIGURES A THRU P DO NOT REPRESENT DETAILS OF CONSTRUCTION THEY SCHEMATICALLY ILLUSTRATE LOCKING CONFIGURATIONS & MOM POSITIONS

CANCELED MS PART NUMBER 1/	SUBSTITUTE PART NUMBER 1/	AVAILABLE LOCKING COMBINATIONS	CIRCUITS MADE WITH TOGGLE LEVER IN KEYING SIDE 1-2 4-5	CENTER POSITION	OPPOSITE KEYING SIDE 2-3 5-6
MS27721-21	MS27721-21-1	ALL	ON	OFF	ON
-22	-22-1	D,F,G	OFF	NONE	ON
-23	-23-1	D,F,G	ON	NONE	ON
-26	-26-1	F	MOM-ON	NONE	ON
-27	-27-1	E,L,N	MOM-ON	OFF	MOM-ON
-31	-31-1	E,F,K,L,M,N	MOM-ON	OFF	ON

1/ The canceled part numbers are no longer procurable; the substitute part numbers have a lower dielectric withstanding voltage at reduced barometric pressure (see 2 below).

(G) denotes changes

APPROVED 6 MAY 1968 REVISED (A) 16 SEPT 1969 (B) 5 AUG 1971 (C) 22 MAY 1972 (D) 14 MAY 1974 (E) 14 AUG 1984 (F) 16 FEB 1987 (G) 20 MAY 1988

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications for repetitive use shall be made from this document when applicable.

USER ACTIVITIES

REVIEWER ACTIVITIES: Air Force 99, Army - AV, Navy - , DLA - ES

PREPARING ACTIVITY: Air Force 85	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Custodians: Navy - AS, Army - ER		SWITCH, TOGGLE, MINIATURE, DOUBLE POLE, TOGGLE SEAL, LEVER LOCK	MS27721
PROCUREMENT SPECIFICATION: MIL-S-83731	SUPERSEDES:		SHEET 1 OF 2

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NOTES:

1. All switches on this standard are designed so that the movement of the switch mechanism is opposite to that of the toggle lever.
2. Test voltage for dielectric withstanding voltage at reduced barometric pressure, shall be 400 V rms minimum.
3. Electrical endurance shall be performed for 30,000 cycles.
4. Mechanical endurance shall be performed 50,000 cycles at -65°C and then 50,000 cycles at 85°C.
5. Intermediate current shall be performed for 30,000 cycles.
6. Terminals shall be molded into the switch case and shall accommodate no. 18 awg wire.
7. Locking arrangement. Positive locking shall be accomplished and shall prevent motion of the toggle lever until the locking mechanism is manually released.
8. The force required to release the locking mechanism shall be 3 to 5 pounds.
9. The locking means at the top of the toggle bushing shall be capable of withstanding a torque of 20 lb. inch applied in both directions immediately following the humidity test.
10. Maximum weight shall be .045 lb.
11. Electrical ratings for 28 volts dc and 115 volts ac at 400 Hz. Lamp load = 1 ampere, resistive load = 5 amperes, inductive load = 2 amperes.
12. Part number example: MS27721-21-1A (locking combination 'A').
13. (G) In the event of a conflict between the text of this standard and the reference cited herein, the text of this standard shall take precedence.
14. (G) Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

USER ACTIVITIES

REVIEWER ACTIVITIES
Air Force 99
Army - AV
Navy
DLA - ES

APPROVED 6 MAY 1968 REVISED (G) FOR CHANGES SEE PAGES 1 AND2

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P A	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Air Force 85 Other Cust Navy - AS Army - ER		SWITCH, TOGGLE, MINIATURE, DOUBLE POLE, TOGGLE SEAL, LEVER LOCK	MS27721
PROCUREMENT SPECIFICATION MIL-S-83731	SUPERSEDES:		SHEET 2 OF 2

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